ReCon1M: A Large-Scale Benchmark Dataset for Relation Comprehension in Remote Sensing Imagery

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***Abstract*— Scene graph generation (SGG) is a high-level visual understanding and reasoning task aimed at extracting entities (such as objects) and their interrelationships from images. Significant progress has been made in the study of SGG in natural images in recent years, but its exploration in the domain of remote sensing images remains very limited. The complex characteristics of remote sensing images necessitate higher time and manual interpretation costs for annotation compared to natural images. The lack of a large-scale public SGG benchmark is a major impediment to the advancement of SGG-related research in aerial imagery. In this article, we introduce the first publicly available large-scale, million-level relation dataset in the field of remote sensing images, which is named ReCon1M. Specifically, our dataset is built upon FAIR1M and comprises 22 262 images. It includes annotations for 873 761 object bounding boxes across 60 categories and 1 052 223 relation triplets across 59 categories based on these bounding boxes. We provide a detailed description of the dataset's characteristics and statistical information. In addition, an efficient global context-aware network (EGCAN) is proposed to improve inference efficiency in dense relation prediction through an object-pair pre-screening mechanism. By integrating visual, spatial, and semantic features, EGCAN captures fine-grained pairwise features and object-level contextual information to enhance its ability to discriminate relation. We conduct two object detection tasks and three subtasks within SGG on this dataset, assessing the performance of mainstream methods on these tasks. The experimental results show that the proposed EGCAN achieves state-of-the-art (SOTA) performance in 17 out of 24 accuracy metrics across three tasks and delivers the best performance in frames per second (FPS) for model inference. The ReCon1M dataset and related resources are available at https://recon1m-dataset.github.io/**

***Index Terms*— Benchmark dataset, relation comprehension, remote sensing, scene graph generation (SGG).**

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I. Introduction

THERE exists a vast number of objects across the Earth, forming a complex interconnected system rather than existing as isolated entities. These objects engage in diverse relations, such as explicit spatial relations and implicit semantic relations. Accurate object relation comprehension bridges the gap between object detection and inference. It is a critical cognitive process that involves knowing, learning, and understanding objects. However, it is challenging to fully capture some relations by shooting at a horizontal angle in daily life because of wide-span relations (e.g., the imminent return of a ship and its corresponding port) and complex relations of multiple objects (e.g., the uninterrupted interplay among thousands of objects within a city). Remote sensing techniques offer a solution, enabling the acquisition of comprehensive, high-resolution surface data from a bird's-eye perspective. This facilitates the observation of diverse objects and provides a solid data foundation for relation comprehension.

Currently, numerous researchers utilize remote sensing data for various object perception tasks, including classification [1], [2], detection [3], [4], and segmentation [5], [6]. The advent of artificial intelligence technology, particularly deep learning, has led to significant breakthroughs in this domain. Models based on remote sensing data can now proficiently recognize multiple objects within images, resulting in substantial improvements in accuracy [7]. Nevertheless, the majority of remote sensing models remain at the level of identifying objects, lacking the capability to understand the relations between objects, which is insufficient for advanced remote sensing cognitive tasks. One primary contributing factor is the scarcity of large-scale, diverse remote sensing relation comprehension datasets.

A large-scale object relation dataset would greatly improve the generalization and accuracy of object relation comprehension models. For instance, in the field of computer vision,

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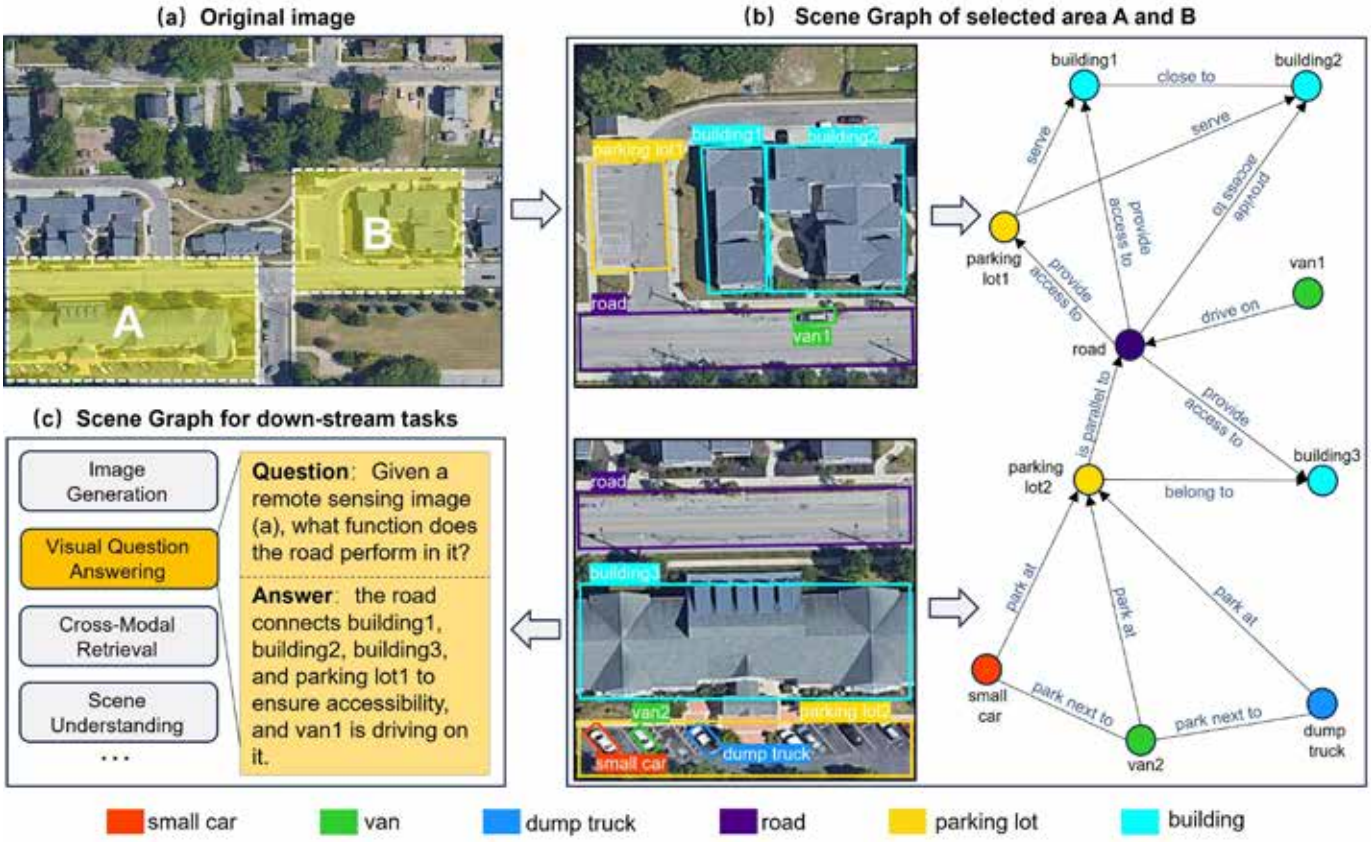


Fig. 1. Visualization example of the ReCon1M dataset. (a) Visualization of the OBB annotations. (b) Due to the complexity of the scene and the large number of objects and related instances, we select regions A and B for scene graph visualization. (c) Several downstream tasks that the scene graph can assist and give a specific example of higher order reasoning in visual question answering.

the emergence of numerous object relation comprehension models has been driven by renowned large-scale object relation datasets such as Visual Genome (VG) [8] and visual relationship detection (VRD) [9]. These datasets leverage natural images captured from a horizontal perspective to facilitate the understanding of object relations. Similarly, to achieve a more profound level of cognitive understanding in remote sensing, there is a growing need for next-generation datasets to serve as training resources and benchmarks for object relation tasks specific to remote sensing applications.

As a result, in order to further advance research in remote sensing relation comprehension study, the following areas related to datasets require further attention and improvements.

1) The size of the datasets needs to be scaled up in terms of the number of images and relation annotations. Scene graph models often have a large number of parameters. To effectively tune these parameters and capture complex relations in data, a large dataset is needed. Without sufficient data, the model may not generalize well to unseen examples. Moreover, models learn hierarchical representations of features from raw data. To capture diverse and nuanced features, a large dataset is necessary to provide enough variation and examples. In general, more data lead to better performance, up to a certain point. As the dataset size increases, the model has more opportunities to learn and improve its performance on the given task.
2) The number of relation categories needs to be expanded. Complex scenes often involve a wide range of interactions and relations between objects. Having more relation types enables the model to adapt to these complexities and accurately represent the scene's content. More relation types allow for a detailed and nuanced representation of the interactions between objects in a scene. By incorporating diverse relation types, the model can generalize better to different scenes and scenarios. This helps improve the model's robustness and performance across a wide range of tasks and datasets, thereby benefiting various applications in computer vision and scene understanding.
3) The importance of semantic relations should be emphasized. Spatial relations hold importance in tasks such as object localization, scene layout analysis, and certain visual reasoning tasks. In contrast, semantic relations provide a higher level understanding of scenes, such as object categorization, scene understanding, and question answering. Semantic relations capture the interactions and dependencies between objects based on their semantic meaning, such as "serve," "cooperate with," "belong to," or "power." These relations provide crucial information about the functional and conceptual aspects of the

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scene, enabling a deeper understanding of the scene's content.

As a result, in order to resolve the issues mentioned above, we propose a novel benchmark dataset for object relation recognition in the remote sensing field, which is named ReCon1M. Some representative examples are given in Fig. 1. We annotate relations in the form of relation triplets ⟨subject, relation, object⟩, where each triplet comprises the coordinates and category of a subject bounding box, the coordinates and category of an object bounding box, and the relation category between them. By representing objects as nodes and relations as edges, a scene graph can be constructed for each image. Based on these annotations, relations between objects in remote sensing images can be explicitly expressed as relation triplets or scene graphs and implicitly represented through deep features extracted by the model for relation classification. This highlights the capability of ReCon1M to advance research related to relation comprehension in remote sensing scenes. To directly and quantitatively assess the "relation comprehension" capability of models, we selected scene graph generation (SGG) as the validation task for our dataset. By understanding relations, models can reason about the roles, functions, and attributes of objects in a scene. This can lead to sophisticated reasoning capabilities, such as understanding causality, temporal relations, and hierarchical structures within the scene. By developing a comprehensive understanding of inter-object relations, models can utilize symbolic reasoning techniques to tackle tasks that demand logical inference and deduction, such as image generation [10], [11], image captioning [12], [13], cross-modal retrieval [14], [15], [16], [17], visual question answering [18], [19], and image understanding and reasoning [20], [21], [22].

In the ReCon1M dataset, the remote sensing imagery comes from the FAIR1M [23] dataset, which is a high-resolution object recognition dataset. Based on the FAIR1M, we label the relations between objects. Since many relations involve new objects that have not been annotated in the original dataset, we annotate these new objects with oriented bounding boxes (OBBs) under the guidance of many experts in remote sensing. To the best of our knowledge, ReCon1M is the largest object relation dataset suitable for remote sensing scenes.

The annotation of relations is a more challenging task than object detection annotation because it is unavoidable that each annotator has recognition bias and sometimes two different annotators give different relation labels for the same image. More seriously, there are unclear boundaries in the annotation of some relations, which makes the task more challenging to accomplish. For example, it is hard to quantitatively define the boundary of relation "close to." Therefore, a unified annotation standard is established to overcome this problem and all annotators follow the same criterion. The standard is consistent during the entire annotation process. As in the above example, we define a rule where if the distance between two objects is less than the longer side of the object and they are not adjacent to each other, then we annotate the relation between these two objects as "close to." In addition to simple spatial relations like "close to," which can be defined using basic spatial computation methods, our dataset encompasses more complex relation categories that require integrated evaluations of spatial relations, object categories, visual context, and surrounding contextual information. These categories vary in difficulty, enabling the assessment of a model's relation prediction capabilities across different levels of complexity. Our goal is for models trained on ReCon1M to develop the ability to understand and discriminate relations across a continuum of types and levels, ranging from simple to complex and from spatial to semantic.

In this article, we introduce a two-stage SGG method, the efficient global context-aware network (EGCAN). EGCAN integrates Rotated Faster R-CNN as an oriented object detector and leverages the related object pairs proposal network (RopPN) for object-pair prefiltering, effectively reducing the number of pairs requiring relation prediction. By utilizing the proposed geometric information embedding context-aware module (GCAM) and object-level context-aware module (OCAM), EGCAN generates fine-grained pairwise features enriched with object-level contextual information, enabling efficient and precise relation prediction.

In summary, the proposed ReCon1M benchmark dataset aims at providing a large-scale object relation dataset for the remote sensing community. With the support of ReCon1M, we hope that a growing number of novel algorithms will emerge in the field of remote sensing image interpretation. The main contribution of this work is summarized as follows.

1) To the best of our knowledge, ReCon1M is the first benchmark dataset in the field of remote sensing imagery with a million-level relation annotation for SGG. The use of OBB annotations in ReCon1M is particularly suited to the characteristics of objects in remote sensing images. This dataset represents a significant advancement over existing datasets in terms of the number and categories of object annotations and the number and categories of relation annotations.
2) We have modified several representative SGG algorithms to accommodate SGG tasks with OBB annotations. We evaluate the performance of these algorithms on our dataset and analyze the results. This evaluation provides a reference benchmark for the design of subsequent SGG algorithms in the field of remote sensing imagery.
3) We propose EGCAN, which leverages the OBB annotations from ReCon1M and employs an oriented object detector to extract discriminative visual features enriched with object orientation information. Through pre-screening object pairs, EGCAN enhances inference efficiency while fully utilizing visual, semantic, and spatial features between objects. By integrating fine-grained pairwise features and object-level contextual information, it achieves effective relation prediction.
4) The experimental results demonstrate that EGCAN achieves state-of-the-art (SOTA) performance in 18 out of 25 metrics across three subtasks. Furthermore, ablation studies confirm the effectiveness of its key modules.

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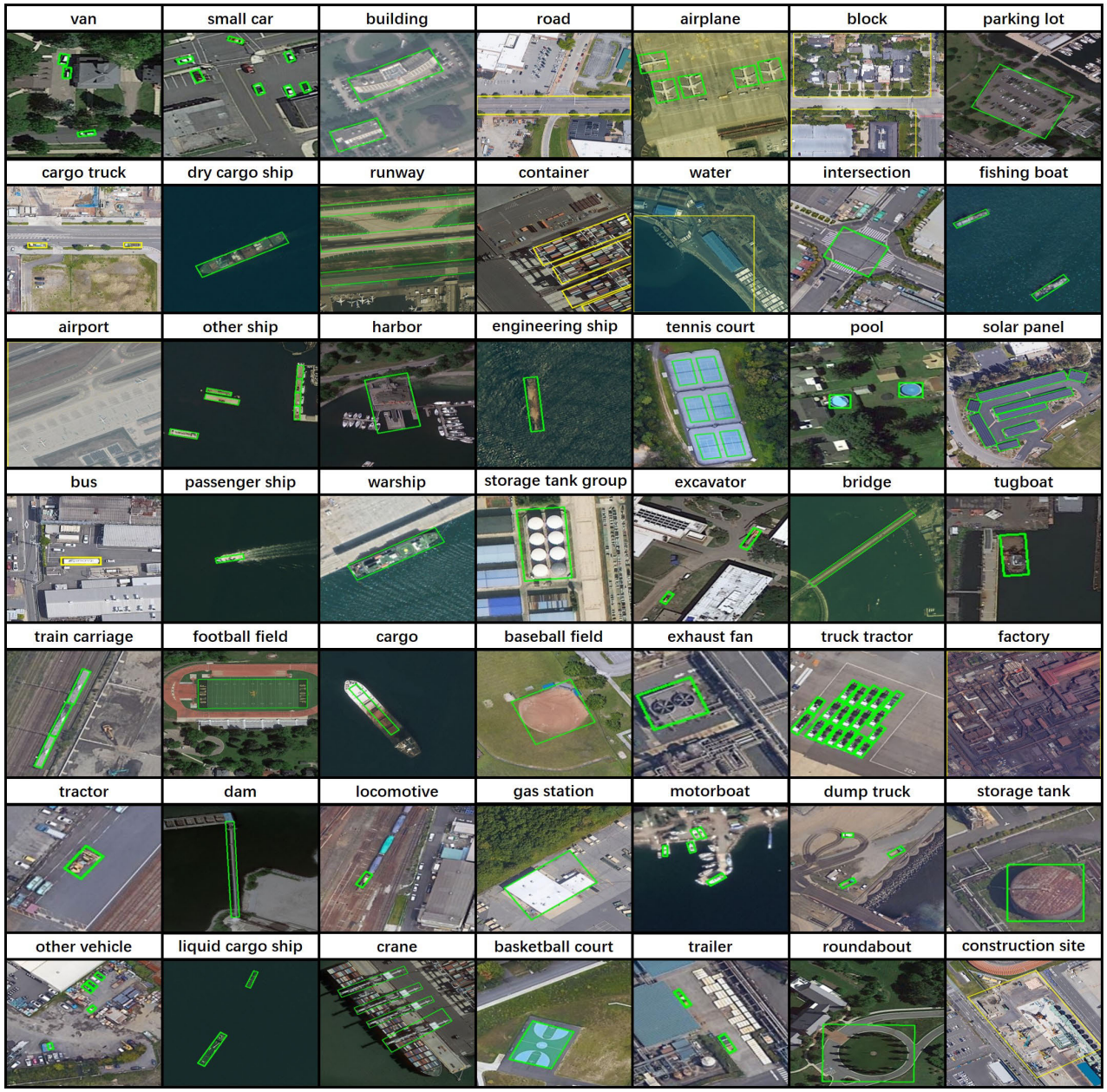


Fig. 2. Example images of some object categories in ReCon1M.

II. Related Work

A. Existing Scene Graph Dataset

Datasets play an important role in the task of SGG. Previously, datasets were predominantly focused on natural scenes. For instance, the VRD dataset [9] comprises approximately 40 000 relation patterns across 100 object categories, encapsulated in five thousand images. Furthermore, the VG [8] has been extensively employed in applications such as visual question answering, captioning, and scene graph construction. It characterizes implicit relations by annotating connections between objects and their attributes, thereby transforming the context into feature-rich data vectors. Notably, Open Images [30] emerges as a substantial dataset, furnishing numerous instances for object detection and relation analysis. Additionally, the RW-SGD [14], constructed based on the Microsoft COCO dataset [31] and YFCC100m [32], facilitates the comprehensive exploration of scene graphs and relation models by manually selecting 5000 images and leveraging Amazon Mechanical Turk (AMT) to generate specific scene configurations. Moreover, the HCVRD dataset [33], an expansion of VRD, includes 1824 object classes, 52 855 images, 927 predicates, and 28 323 relation classes. For the generation of dynamic scene graphs, the CAD120 [9] and Action Genome (AG) [34] collectively provide a video action reasoning dataset that captures human social scenes and focuses on the temporal aspects of relation pattern analysis.

However, datasets based on relation patterns for remote sensing are currently scarce. A comprehensive comparison of commonly used scene graph datasets in both natural and

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Fig. 3. Example instances of some relation categories in ReCon1M. A relation instance can be represented by a triplet ⟨subject, relation, object⟩. Each relation instance is represented by two images, where the left and right images, respectively, show the subject and the object, with the relation label above and the labels of the subject and object below the images.

remote sensing domains is presented in Table I, highlighting the differences in scale, annotation richness, and applicability across domains. The RSICD dataset [35] enables a detailed linguistic representation of remote sensing images. It stands out by providing extensive captioning data, which facilitates the examination of associated knowledge through the relations it generates. Additionally, Li et al. [28] introduced the remote sensing image scene graph dataset (RSSGD). They explored the construction of scene graphs for remote sensing images and incorporated global information using graph convolution networks, enhancing the analysis of spatial and relation data within these images.

B. Existing Scene Graph Methods

There is a lengthy research history behind the concept of building an in-depth understanding network based on relation patterns among objects. Semantic reasoning has been extensively studied in the realm of computer vision [31], [36], [37], [38], many innovative scholars have produced significant discoveries in the field of scene graph research based on semantic reasoning [15], [39]. In the realm of SGG methods, a distinction arises between statistically informed methodologies and those rooted in deep learning.

Conditional random fields (CRFs), a classical tool in this context [40], exhibit proficiency in incorporating relation infor-

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TABLE I
STATISTICS OF THE SCENE GRAPH DATASET IN BOTH COMMON SCENE AND REMOTE SENSING SCENE. "−" INDICATES THAT THIS ATTRIBUTE IS NOT RELEASED

Scene	Dataset	Size	Image size	Objects			Relations		Year
				#annotation	#bbox	#categories	#triplet	#categories	
common	Visual Phrase [24]	2,769	100∼801	HBB	3,271	8	1,796	17	2011
	RW-SGD [14]	5,000	-	HBB	93,832	6,745	112,707	1,310	2015
	VRD [9]	5,000	192∼1280	HBB	90,000	100	37,993	6,672	2016
	Visual Genome [8]	108,077	72∼1280	HBB	3,843,636	33,877	2,347,187	42,374	2017
	VrR-VG [25]	58,983	72∼1280	HBB	282,460	1,600	203,375	117	2019
	Open Images v4 [26]	9,278,275	∼1024	HBB	15,440,132	600	391,073	329	2020
	UnRel [27]	1,000	-	HBB	-	-	76	18	2020
remote sensing	RSSGD [28]	-	-	HBB	40,000	39	-	13	2021
	S2SG [29]	-	256	MASK	-	-	1200	12	2022
	ReCon1M(ours)	22,262	400∼10,000	OBB	873,761	60	1,052,223	59	2024

mation into the models. In the SGG domain, CRF methods include two fundamental components: object detection and relation detection. Several refined SGG models, leveraging CRF principles, have showcased improved performance through the integration of robust relation prediction models and object detection models such as the deep relationship network (DR-Net) [41] and the semantic compatibility network (SCN) [42]. However, there is a tendency to overlook the inherent order of relationships between two entities, leading to the confusion of subjects and objects. This limitation is particularly pronounced in tasks involving directed knowledge graph prediction, contributing to notable distortions in knowledge acquisition for relation prediction. Knowledge graphs, representing and analyzing information in scene graphs, manifest complex structures analogous to vector graphs, wherein relations are distinctly defined through sets of entities and edges. Models grounded in translation embedding (TransE) have demonstrated exceptional utility in this context [43]. TransE conceptualizes relations as translations, treating the relation between a pair of objects as the translation of the attribute vector from one object to another, thereby acquiring optimal translation patterns. Drawing inspiration from TransE's success in knowledge learning, recent investigations [43], [44], [45] explore the generation of visual connections through object features. Furthermore, methodologies like translation-based model with dynamic mapping matrix (TransD) [46] and analogy transfer (AT) [47] simplify the knowledge vectors in scene graphs through transformations. However, this approach only focuses on the relation features between subjects and objects, neglecting the integration of contextual information. Consequently, numerous unresolved challenges persist within current TransE methodologies.

Beyond methodological challenges, there are significant defects in training data. Notably, the sparsity issue in predicate representations poses challenges, similar to the long-tail effect, complicating the task of visual relation detection. Using the Stanford VRD dataset [9] as an illustrative case, its potential predefined relations count exceeds 700k, highlighting that numerous real relations lack any training samples. UVTransE [48] introduces joint features for subject and object boundaries, effectively capturing contextual information. This class of AT-based SGG methods exhibits promising applications, and empirical findings substantiate their effectiveness.

C. Limitation and Challenges in SGG

1) Long-Tail Problem in SGG: Within the context of existing scene graphs, extant methodologies for relation recognition exhibit limitations in effectively addressing the prevalent long-tail problem. This issue arises from a nonuniform distribution of samples corresponding to identified relation patterns, leading to a pronounced concentration in the head and a flattened distribution in the tail. Consequently, most of the relation patterns in the training dataset suffer from an inadequacy of supporting data. The complexity of relations amplifies exponentially in scenarios involving multiple relations, and the reality of many visual relations in authentic scenarios being absent from training sets poses a challenge in achieving comprehensive cognition.

To overcome this challenge, current research in scene graph analysis endeavors to mitigate the consequences of the long-tail distribution. Noteworthy among these approaches are zero-shot and few-shot learning methodologies [49], [50], [51], [52], [53], strategically transferring relation knowledge to address long-tail positions, just like the principles of meta-learning. This strategic transfer enhances the model's capacity to learn from limited samples. Furthermore, the integration of prior knowledge methods introduces external information into the model. Specifically, introducing linguistic priors [9], [54], [55] and statistical priors [41], [56], [57], [58] serves to improve relation recognition accuracy. The alignment with prior knowledge contributes to an augmented scalability and heightened recognition capabilities of the model.

Moreover, from the standpoint of transfer learning, redistributing head relations to the tail emerges as a potential strategy for dealing with the long-tail effect [59]. The total direct effect (TDE) explores a training paradigm rooted in causal reasoning, where a comparative analysis of the discovered causal relation network and the counterfactual network derived from the scene graph aims to identify and correct errors in the transfer learning process [60].

Despite previous datasets such as Open Images, Visual Phrase (VP), RW-SGD and HCVRD [14], [24], [30], [33]

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TABLE II
NUMBER OF INSTANCES FOR EACH OBJECT CATEGORY AND DATASET SPLIT

Category	Train	Validation	Test
van	89424	29617	59677
small car	84045	27204	56743
building	68653	23354	46939
road	36139	11673	24176
airplane	17565	6099	11579
block	14727	4962	10577
parking lot	14503	4884	9779
motorboat	12505	5158	10350
dump truck	13432	4358	8848
cargo truck	8826	3174	5764
dry cargo ship	8106	2260	5502
runway	7168	2516	4577
container	6583	2088	4782
water	4963	1669	3296
intersection	4787	1507	3343
fishing boat	4007	1112	2463
other vehicle	3918	1305	2211
storage tank	3560	920	2228
airport	3185	1072	2074
other ship	2467	812	2039
harbor	2539	793	1730
solar panel	1713	673	1219
pool	1801	565	1179
tennis court	1821	522	1121
engineering ship	1777	459	1217
crane	1534	420	1011
liquid cargo ship	1420	418	961
bus	1132	306	735
passenger ship	977	311	818
warship	1088	365	616
excavator	956	282	517
storage tank group	754	413	541
bridge	779	259	656
basketball court	726	208	512
trailer	771	217	439
tugboat	748	219	457
train carriage	798	233	284
football field	582	224	416
cargo	627	136	410
baseball field	585	187	393
boarding bridge	549	193	314
greenbelt	495	211	346
exhaust fan	414	140	301
truck tractor	259	260	241
factory	312	107	240
construction site	321	121	214
roundabout	281	80	140
terminal	167	43	98
tractor	165	49	72
railway	70	31	62
farmland	63	22	32
stadium	57	15	40
chimney	25	17	20
gas station	23	10	28
dam	31	8	19
locomotive	33	6	17
expressway service area	19	10	25
control tower	30	7	15
smoke	20	16	14
helicopter apron	10	2	7

TABLE III
NUMBER OF INSTANCES FOR EACH RELATION CATEGORY AND DATASET SPLIT

Category	Train	Validation	Test
parked at	103882	34025	68871
park next to	99026	32460	64643
close to	75645	25453	54491
provide access to	69107	23230	47383
inside	49685	16143	34061
drive on	32512	10676	21622
moor at	22567	7863	17194
serve	11763	4033	7330
is parallel to	10967	3449	7504
adjacent to	8412	2505	5676
belong to	6693	2358	4348
sail on	6731	2194	4328
pile up at	4802	1571	3758
cross	4128	1201	2726
supplement	2183	787	1368
slow down	1762	500	1135
taxi on	1550	559	977
supply	1508	280	972
cooperate with	1287	274	791
contain	807	424	514
power	664	302	482
link to	708	201	213
prepared for	480	159	445
support	577	194	297
hoist	544	149	368
above	504	152	392
drive at the different lane	628	167	219
dock at	568	179	266
connect	508	192	309
drive at the same lane	621	228	154
border	461	190	337
equipped with	502	162	285
separate	226	106	169
ventilate	247	76	175
transport	252	54	160
support the construction of	227	87	134
manage	231	78	122
placed on	170	78	122
sail by	72	103	119
lie under	126	26	80
park alone at	95	21	42
cultivate	73	23	36
converge	89	19	23
tow	62	10	38
provide shuttle service to	53	19	35
around	52	14	37
move away from	45	21	34
exit from	38	17	27
enter	34	16	26
adjoint with	32	14	9
dock alone at	26	6	22
load	25	9	18
command	30	7	15
is symmetric with	18	10	22
block	27	5	15
emit	19	16	11
pass under	22	3	19
dig	20	8	15
pull	25	4	11

contain a wide range of relation classes, they also exhibit a long-tail distribution of infrequent relations, sharing the same issue of sparse data in their long-tail region.

2) Challenges in Remote Sensing Image Datasets: Previous remote sensing datasets, including DIOR, HRRSD, and NWPU [61], [62], [63], lack annotations for certain relation patterns, which makes it harder to model and compute the key relation in relation study. Therefore, creating a useful dataset based on relation patterns becomes essential when considering the features of remote sensing images and the complexity of relations. Relation-pattern-based remote sensing datasets are now very few. Because the spatial relations in the RSICD dataset [35] are not well-represented in image captions, it is quite difficult to directly use this dataset for modeling and calculation. Semantic relation model and dataset for remote sensing scene understanding fail to thoroughly

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analyze the characteristics of remote sensing images, and the feature relations in the dataset itself are insufficient, unable to cope with large-scale effective relation studies.

Drawing inspiration from TransE's success in knowledge learning, recent investigations [43], [44], [45] explore the generation of visual connections through object features. Furthermore, methodologies like RLSV [46] and AT [47] simplify the knowledge vectors in scene graphs through transformations. However, this approach only focuses on the relation features between subjects and objects, neglecting the integration of contextual information. Consequently, numerous unresolved challenges persist within current TransE methodologies.

III. Details of Dataset

This section details the dataset, including its source and preprocessing, object and relation categories, and annotation methods.

A. Images Sources and Dataset Splits

To efficiently construct a large-scale remote sensing relation dataset annotated with OBB, we selected the FAIR1M dataset as our foundation. FAIR1M is a large-scale standard benchmark dataset used for fine-grained object detection and recognition tasks in high-resolution remote sensing images. It contains over one million instances and more than 40 000 images collected from high-resolution satellites and Google Earth, with resolutions ranging from 0.3 to 0.8 m. FAIR1M employs OBB annotations, categorizing objects into five major classes and 37 fine-grained categories. We use images from FAIR1M and Google Earth as our source, selecting 22 262 images for additional object bounding box and relation annotations.

For the dataset split, we use half of the images for the training set, one-sixth for the validation set, and one-third for the test set. The dataset includes both the original images and their corresponding OBB and relation annotations. Detailed split information for the objects and relations is shown in Tables II and III.

B. Category Design

The original annotations of object categories in FAIR1M did not meet our requirements for constructing a large-scale remote sensing relation dataset. Therefore, while retaining the original FAIR1M annotations, we conduct extensive reannotations to expand the object categories and add a substantial number of relation annotations.

1) Objects Category: FAIR1M includes five major categories and 37 fine-grained object categories, with the fine-grained categories containing richer semantic information. We chose to retain these 37 fine-grained categories as the true labels for the original bounding boxes. However, since these 37 categories only belong to five major classes, the relation information they contain is still not sufficiently rich. Hence, there is a need to expand the object categories. Each category could have semantic or spatial relations with others, so increasing the number of categories significantly enhances the richness of the relation information. We retain the "airplane" category from FAIR1M along with other 26 fine-grained categories (passenger ship, bridge, motorboat, fishing boat, tugboat, intersection, and so on) and additionally add 34 new categories: building, road, block, parking lot, runway, container, water, storage tank, airport, harbor, solar panel, pool, crane, storage tank group, train carriage, cargo, boarding bridge, greenbelt, exhaust fan, factory, construction site, terminal, railway, farmland, stadium, chimney, gas station, dam, locomotive, expressway service area, control tower, smoke, and helicopter apron. These categories are selected by remote sensing image interpretation experts based on their prevalence, relation information, and practical value in applications. Typical objects corresponding to each category are illustrated in Fig. 2.

2) Relation Category: After in-depth discussions with professional remote sensing image interpretation experts, we identify 59 common and practically valuable relation categories in remote sensing scenarios. These can be divided into two major categories based on their focus: spatial relations and semantic relations. Spatial relations, which focus on the spatial distribution relations of objects, include 26 types: parked at, park next to, close to, inside, is parallel to, adjacent to, moor at, sail on, sail by, pile up at, taxi on, drive on, dock at, dock alone at, link to, cross, pass under, above, separate, placed on, park alone at, drive at different lane, drive at the same lane, lie under, around, and is symmetric with. Semantic relations, on the other hand, emphasize advanced semantic relations between objects and include 33 types: provide access to, serve, belong to, supplement, slow down, supply, cooperate with, contain, power, prepared for, support, hoist, connect, border, equipped with, ventilate, transport, support the construction of, manage, cultivate, converge, tow, provide shuttle service to, move away from, exit from, enter, adjoint with, load, command, block, emit, dig, and pull. Typical diagrams for each type of relation are illustrated in Fig. 3.

C. Image Annotaion

1) Annotation Format: Our relation dataset requires two types of annotations: object annotations and relation annotations. Common relation datasets in natural scenes, such as VG and HVCRD, use horizontal bounding boxes (HBBs) without directional information for object annotations. This is because their images come from natural scenes, where the images are smaller, the objects are simpler, and there are fewer distracting elements. However, the span of remote sensing images is much broader than that of natural scenes, and their object complexity is significantly higher. Using HBB for annotations in remote sensing could compromise the precision of the annotations. For example, if a road diagonally crosses the entire image, using HBB would mean the size of the bounding box is the size of the entire image, with foreground pixels significantly outnumbered by background pixels. This leads to inaccurate object information extraction, thus affecting the accuracy of relation prediction between objects.

Using OBB $\{(x_i, y_i), i = 1, 2, 3, 4\}$, where (x_i, y_i) represents the vertices of the bounding box, addresses these issues effectively. For objects in remote sensing images that

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(a) (b)

Fig. 4. Comparison of bounding box annotations in the same source images between FAIR1M and ReCon1M. (a) Annotations of FAIR1M. (b) Annotations of ReCon1M.

have a clear forward orientation, such as airplanes, ships, and vehicles, we choose the vertex on the front left as the first vertex of the OBB and annotate the other three vertices clockwise. For objects without a clear forward orientation, such as oil tanks, containers, and pools, we select the top-left vertex as the first vertex and mark the remaining three vertices clockwise. For relation annotations, similar to most natural scene relation datasets, we use the ⟨subject, relation, object⟩ triplet format. Each subject and object is one of the previously mentioned 60 object categories, and each relation is one of the 59 relation categories mentioned earlier.

2) Annotation Procedure: We develop a specialized annotation tool to add relation annotations between object bounding boxes and develop a dedicated visualization tool for visualizing relation annotations in each image. To ensure the accuracy of the dataset annotations, we establish a comprehensive annotation process for the remote sensing relation dataset.

1) *Discussion With Experts:* We engage with professional remote sensing image interpreters to select object and relation categories that are common and have practical application value based on the existing dataset images and real-world scenarios.
2) *Example Selection and Annotation Guidelines:* Professionals select typical examples for each object and relation category from the images to be annotated. They provide specific guidelines and examples for each category to ensure consistency and accuracy.
3) *Standardization of Annotation Manual:* A standardized annotation manual is written, and professionals instruct the annotators on the standards and important considerations required for accurate annotations.
4) *Dual Annotation Teams:* The annotators are divided into two groups, A and B. Both groups concurrently perform object annotations and relation annotations. After completing the annotations, each group uses the visualization tool to check the accuracy of the other group's annotations.
5) *Quality Control and Review:* Approximately 20% of the annotated data are randomly selected and reviewed by professionals. If any issues are identified, feedback is provided to the annotators for correction, and this process is repeated until the professionals confirm the annotations are accurate.

This rigorous process ensures that our dataset not only meets high standards of accuracy but also is useful for developing and training models in remote sensing image interpretation and relation analysis.

D. Dataset Statistics

The input data and annotation formats for this dataset have already been described in Section III. Considering the characteristics of this dataset, we conduct two object detection tasks and one relation detection task on it. The object detection tasks include HBB detection and OBB detection. After performing object detection, several object bounding boxes and their categories will be obtained. In the relation detection task, for a given input image, we output several relation triplet instances ⟨subject, relation, object⟩, where both subject and object include the coordinates and categories of the bounding boxes, and relation represents the association description between subject and object.

1) Instance: ReCon1M includes 873 761 object instances with category annotations. We have added a large number of dense object annotations based on FAIR1M. Fig. 4 illustrates the difference in the number of object annotations between FAIR1M and ReCon1M on the same image. As shown in Table II, there is a significant disparity in the number of instances across different categories. For example, the "van" category has the highest number of instances at 178 718, accounting for 20.4539% of total object instances. In contrast, the "solar panel" category, which is at the median, has 3605 instances, making up 0.4126% of the total. The "helicopter apron" category has the fewest instances at only 19, representing only 0.0022% of the total. This is a typical example of class imbalance in remote sensing image datasets, where some categories far outnumber others. This imbalance reflects real-world remote sensing scenarios and adds to the challenges of our dataset. Fig. 5 also shows the distribution of the number of instances per image, with up to 1200 instances in some images and an average of 39 per image, which is a significant increase compared to previous SGG datasets.

Remote sensing images cover a wide area with a large variance in object scale sizes, posing significant challenges

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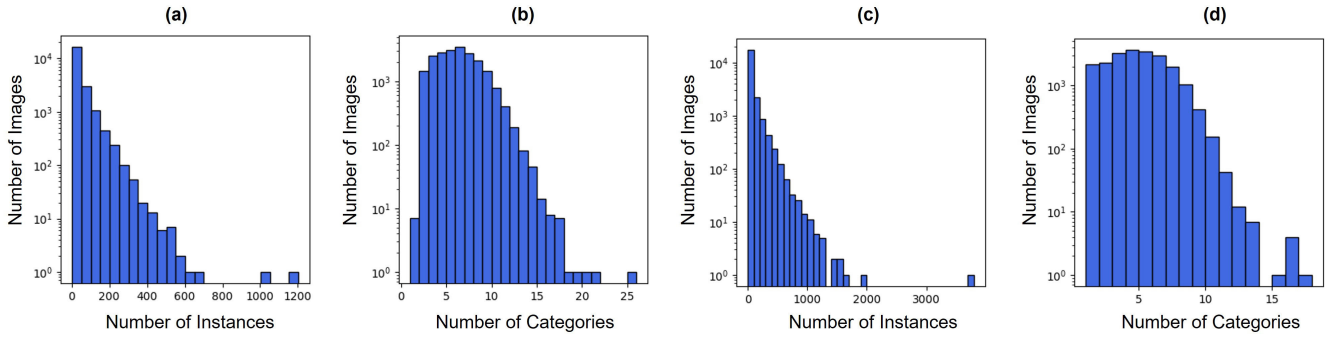


Fig. 5. (a) Distribution of the number of object instances per image. (b) Distribution of the number of object categories per image. (c) Distribution of the number of relation instances per image. (d) Distribution of the number of relation categories per image.

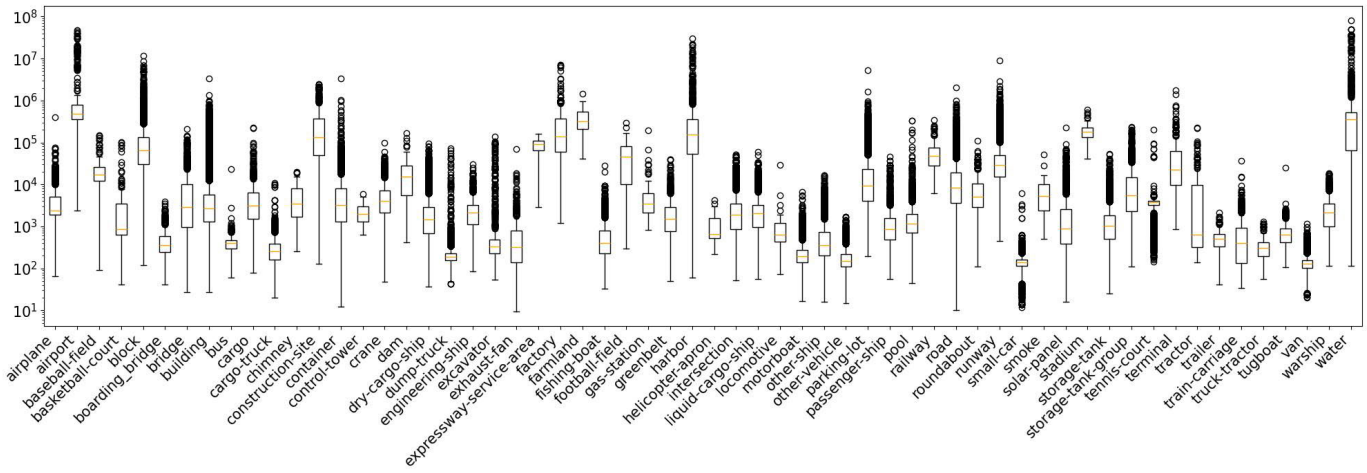


Fig. 6. Size variations for each object category in ReCon1M. The sizes of different categories vary in different ranges.

for object detection and relation recognition. We classify the objects into three sizes based on the area (bounding box size): large objects with pixel counts of 2048 or more, medium objects with pixel counts between 144 and 2047, and small objects with pixel counts between 11 and 143. The statistical results show that the proportions of large, medium, and small objects are 34.9%, 38.0%, and 27.1% of the total number of objects, respectively. The relation between object scale and category is also significant, as shown in Fig. 6. For example, the average area of category "parking lot" is 22305.7, whereas for category "van," it is only 132.8. There are also large variations in size within the same category; for example, the largest area in category "building" is 3 315 169, and the smallest is 27.

2) Relation: Relations are the core component of our dataset, represented as relation triplets ⟨subject, relation, object⟩, which we refer to as relation instances. Our dataset contains 1 052 223 relation annotations. Fig. 7 shows the number of relation instances for each relation category. It is evident that the distribution of relations exhibits a typical long-tail pattern. The most frequent relation category is "parked at," with 206 778 instances, accounting for 19.6512% of all relation instances, while the least frequent is "pull," with only 40 instances, or 0.0038% of the total. Like the object categories, there is a significant imbalance among the relation categories. Additionally, the distribution of relations is biased and often exhibits a strong correlation with the object categories, as demonstrated in Fig. 8. The bias in the dataset reflects real-world phenomena, for example, boats are mostly seen sailing on water rather than cars, and airplanes taxi on runways rather than ships, whereas ships do not. The data bias is a critical issue in real-world applications, lending significant relevance and added challenge to our dataset.

IV. Method

The baseline methods we adopt are designed for relation prediction tasks based on HBB in natural scenes, whereas ReCon1M is a dataset annotated with OBB and used in remote sensing scenes. Consequently, directly replacing the detector in the baseline methods with a rotated detector does not yield satisfactory results. Therefore, to address the issues of OBB annotations, dense objects and relations, and large variations in object scales and distances between objects in ReCon1M, we propose an EGCAN, which is inspired by the attention mechanism employed in [64] for object feature interaction and fusion. The overall structure is shown in Fig. 9. It mainly consists of four stages: oriented object detection, related object-pair proposal, pairwise feature extraction, and relation triplet prediction. The pairwise feature extraction

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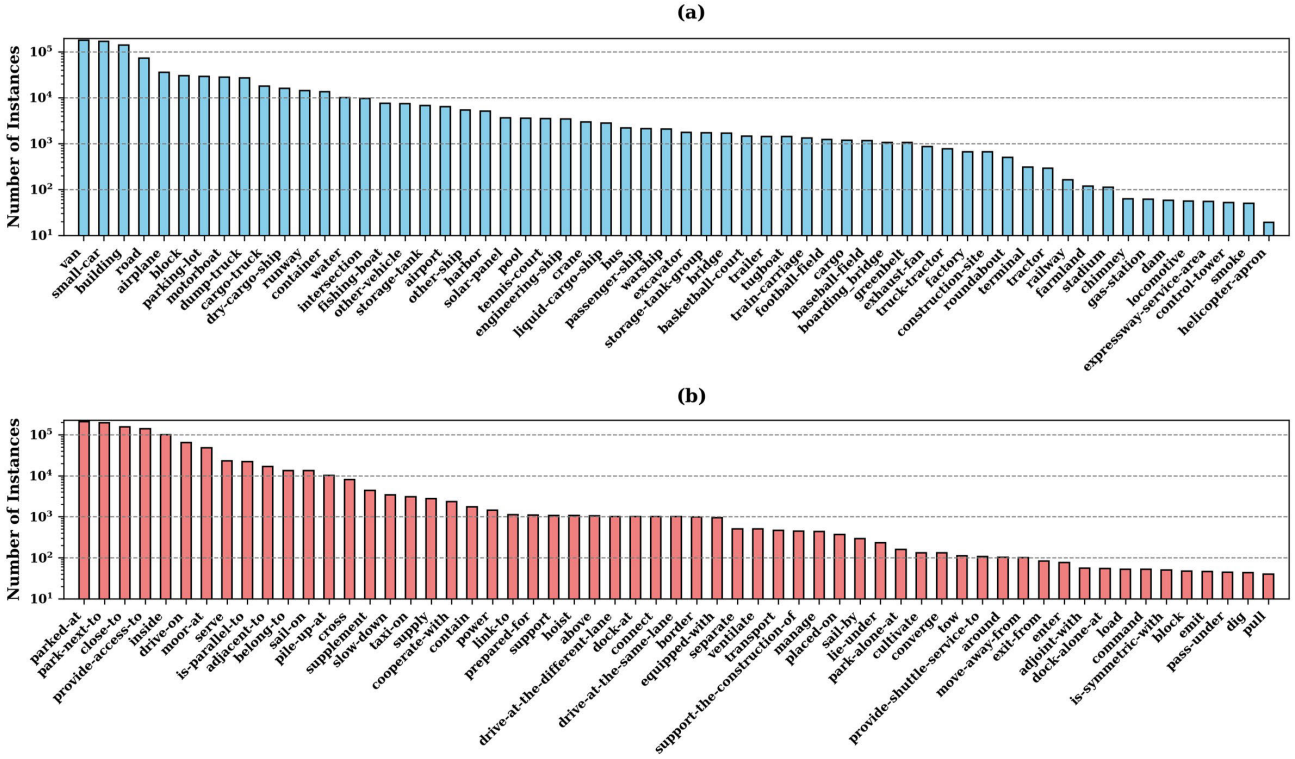


Fig. 7. (a) Distribution of the number of instances per object category. (b) Distribution of the number of instances per relation category.



Fig. 8. Diagram of correlation strength between some object categories. Each grid point's value represents the logarithm of the number of relations occurring between two object categories. The darker the color, the higher the correlation strength between the two object categories, and vice versa.

process includes a GCAM and an OCAM. For the object information and features output by the detector, the RopPN is used for pre-screening object pairs to improve the efficiency of relation prediction. Subsequently, pairwise feature extraction is conducted to obtain global context-aware features with geometric information embedding for each object pair to be classified. Finally, relation prediction is performed on each object pair using the relation triplet prediction module (RTPM).

A. Object Detection

Considering that most two-stage SGG models previously used Faster R-CNN as the object detector, we adopt Rotated Faster R-CNN as the OBB detector. The main difference between Rotated Faster R-CNN and Faster R-CNN is that the former can detect objects at any angle of rotation. It achieves this through an improved region proposal network (RPN), using rotated candidate regions and adding an angle parameter in the bounding box regression, making it more suitable for OBB detection tasks. In two-stage SGG methods, the relation prediction part relies on the accuracy of the detector and the quality of the extracted visual features. Using Rotated Faster R-CNN, specifically designed for OBB detection, can better apply to the ReCon1M dataset and better support the subsequent relation prediction components. In this stage, the detector will output the set of bounding boxes $\boldsymbol{B} = \{b_i\}_{i=1}^{N_{\text{box}}}$ where $b_i \in \mathbb{R}^{N_{\text{box}} \times 6}$, multiscale feature maps F_{fpn}, and the logits for each box $\boldsymbol{l} \in \mathbb{R}^{N_{\text{box}} \times C_{\text{obj}}}$ where C_{obj} refers to the total number of object categories. The object bounding boxes can extract corresponding visual features $\boldsymbol{f} = \{f_i\}_{i=1}^{N_{\text{box}}}$ from F_{fpn} using RoIAlign, $f_i \in \mathbb{R}^{C_{\text{roi}} \times P_h \times P_w}$. C_{roi} is the number of channels in the feature map, and P_h and P_w are the height and width of the grid into which the ROI is divided during the ROIAlign process, respectively. Here, we employ Rotation-invariant RoI Align (RiROIAlign), which adaptively extracts

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rotation-invariant features from equivariant features according to the orientation of RoI.

B. Related Object Pair Proposal

Theoretically, there are $N_{\text{box}} \times (N_{\text{box}} - 1)$ possible potential relations between N_{box} objects, and one of the biggest differences between ReCon1M and previous natural scenes is that the object and relation annotations are much denser. As the number of objects N increases, the number of potentially related object pairs grows exponentially. During model training, we can solve this problem by sampling no more than the maximum specified number of positive and negative samples. However, during model inference testing, it is necessary to predict the relations between all possible related object pairs output by the object detector, which means that the demand for computing resources and the time taken for model inference both grow exponentially with the number of objects N_{box}, which is unacceptable. To address this challenge, we propose RopPN, which effectively reduces the number of potential object pairs to a manageable subset by predicting relevance scores for object pairs based on their semantic, spatial, and visual features. Specifically, for each object pair (b_i, b_j) detected by the object detector, the RopPN computes a pairwise feature representation that combines semantic embeddings and visual embeddings.

1) Foreground and Background Pair Sampling: RopPN begins by matching object proposals with ground-truth (GT) bounding boxes using intersection over union (IoU). A proposal object pair is considered a positive match if its IoU with a GT box exceeds a threshold τ. Specifically

$$\text{IoU}(b_i, b_j) = \frac{|b_i \cap b_j|}{|b_i \cup b_j|} \tag{1}$$

where b_i and b_j denote the bounding boxes of two objects. Proposals satisfying IoU $> \tau$ are marked as a valid match, where τ is set to 0.5.

During training, RopPN samples foreground and background pairs using a balanced strategy. Let M be a binary matrix denoting matched pairs. Foreground pairs and background pairs are defined as

$$M_{\text{fg}}(i, j) = \begin{cases} 1, & \text{IoU}(b_i, b_{\text{gt}}) > \tau \text{ and IoU}(b_j, b_{\text{gt}}) > \tau \\ 0, & \text{otherwise} \end{cases} \tag{2}$$

$$M_{\text{bg}}(i, j) = 1 - M_{\text{fg}}(i, j). \tag{3}$$

2) Pairwise Feature Extraction for Object Pair Proposal: The object-pair proposal phase serves as the relation prescreening stage, where a binary classification is performed for each object pair to determine whether a relation exists. To reduce computational cost, we extract low-resolution visual features $\tilde{f}_i \in \mathbb{R}^{C_{roi} \times (P_h/2) \times (P_w/2)}$ from F_{fpn}. Each object's feature vector is transformed into subject and object embeddings and is further enhanced using a self-attention mechanism

$$v_i^{\text{sub}} = \text{SelfAttention}(\text{MLP}(\tilde{f}_i)) \tag{4}$$

$$v_i^{\text{obj}} = \text{SelfAttention}(\text{MLP}(\tilde{f}_i)). \tag{5}$$

The object class logits $\boldsymbol{l}$ are transformed into semantic embeddings using the GloVe [65] embedding $\boldsymbol{E} \in \mathbb{R}^{C \times d_s}$ (where d_s is the size of each semantic embedding vector)

$$\boldsymbol{s} = \boldsymbol{p}\boldsymbol{E}. \tag{6}$$

For each object pair (b_i, b_j), the pairwise representation h_{ij} is constructed by concatenating the features

$$h_{ij} = \left[s_i, v_i^{\text{sub}}, s_j, v_j^{\text{obj}}\right]. \tag{7}$$

3) Relevance Scoring: The pairwise feature h_{ij} is passed through a fully connected network to compute the relevance score. The final relevance score r_{ij} is computed as

$$r_{ij} = \sigma(\text{MLP}(h_{ij})) \tag{8}$$

where $\sigma(\cdot)$ is the sigmoid function.

During training, GT object pairs are used to supervise the RopPN. Let. The binary cross-entropy loss is applied as

$$\mathcal{L}_{\text{RopPN}} = -\frac{1}{|\mathcal{P}|} \sum_{(i,j) \in \mathcal{P}} \left[y_{ij} \log(r_{ij}) + (1 - y_{ij}) \log(1 - r_{ij})\right] \tag{9}$$

where $\mathcal{P}$ is the set of all sampled object pairs; $y_{ij} \in \{0, 1\}$ is the GT label for the pair, which indicates whether a relation exists between (b_i, b_j); and $|\mathcal{P}|$ is the total number of pairs.

During inference, we retain the top N object pairs with the highest relevance scores for subsequent relation prediction, where N is a hyperparameter.

C. Pairwise Feature Extraction

The ultimate goal of EGCAN is to predict potentially related object pairs and the latent relations between them. In typical two-stage SGG methods, final relation prediction requires the model to predict the relation category label for each object pair based on the discriminative feature vectors generated. Therefore, extracting more discriminative features at the pairwise level is crucial.

1) Geometric Information Embedding Visual Context-Aware Module: Visual features are critical for predicting object relations. Local visual feature representations of the subject and object can be obtained from the feature map output by the detector. Additionally, global visual feature representations with a larger receptive field can be derived from the minimal enclosing horizontal rectangle of the subject and object bounding boxes. Acquiring the geometric properties of object pairs, such as coordinates, area, scale differences, and distance, can also serve as valuable supplementary information. For this purpose, we propose GCAM, which extracts comprehensive object-pair representations by combining global–local visual context with geometric information. These features are fused to generate a high-quality representation for each object pair, enhancing the model's capability to infer relations accurately.

The visual feature f_i of the subject box b_i, f_j of object box b_j, and f_{ij}^{union} of the smallest enclosing bounding box that encompasses both the subject and object boxes can be extracted from F_{fpn}. We concatenate the visual features of the subject, object, and union region along the channel dimension, fuse the features through two convolutional layers, flatten the

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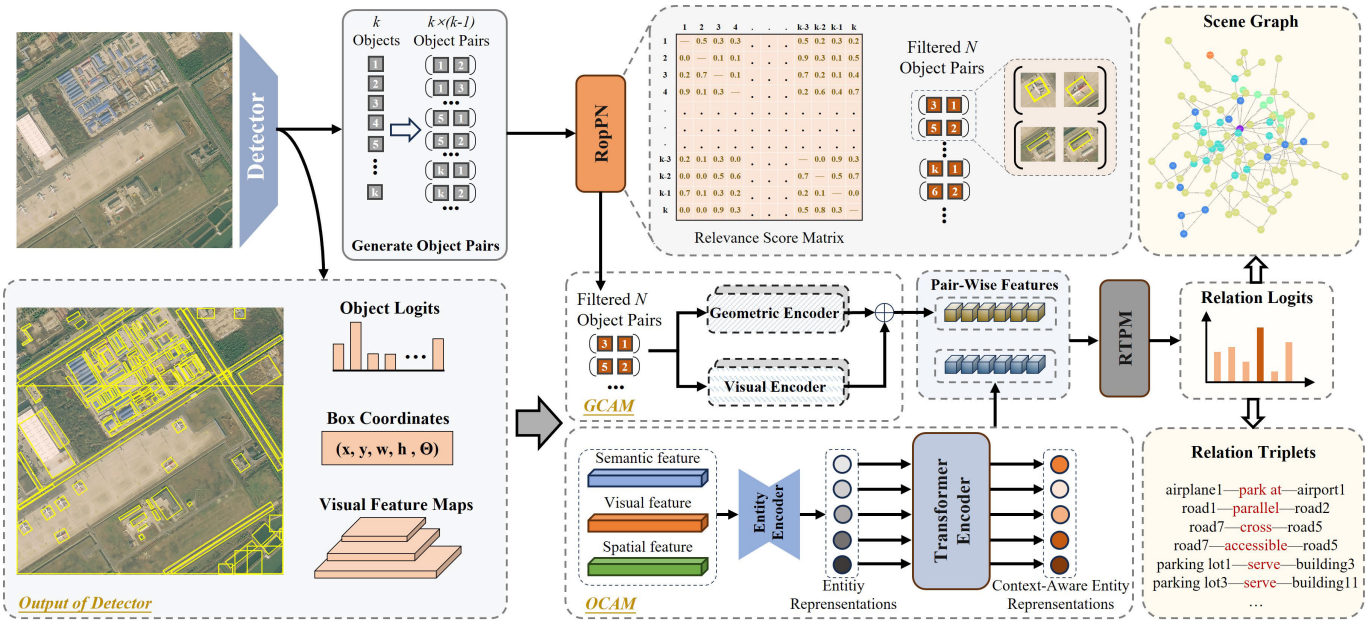


Fig. 9. Overall framework of the proposed EGCAN. After inputting the image into the object detector, the outputs include the bounding box coordinates, object logits, and multiscale visual feature maps, which are used for subsequent relation prediction. For the k objects detected, there may be $k \times (k-1)$ possible object pairs. After feeding all potential object pairs into the RopPN, it generates a $k \times k$ dimensional relevance score matrix and retains the top N object pairs with the highest relevance scores. The GCAM generates pairwise features that are context-aware by leveraging both geometric and joint visual features of the objects. The OCAM combines the semantic, visual, and spatial features of each object for entity encoding. Each object's entity representation is then encoded using a TESA mechanism to obtain context-aware entity representations at the object level. The entity representations of the subject and object are concatenated to produce the corresponding pairwise features. The pairwise features generated by the GCAM and OCAM are concatenated and fed into the RTPM to make the final relation predictions for the N object pairs, producing relation triplets and a scene graph corresponding to the input image.

result, and then use two linear layers to map the features to a low-dimensional space for relation prediction

$$f_{ij}^{\text{conv}} = \text{ReLU}\big(\text{Conv2D}\big(\big[f_i, f_j, f_{ij}^{\text{union}}\big]\big)\big) \tag{10}$$

$$f_{ij}^{\text{visual}} = \text{FC}\big(\text{Flatten}\big(f_{ij}^{\text{conv}}\big)\big). \tag{11}$$

The geometric properties of each object pair (b_i, b_j) are embedded into a 15-D feature vector $g_{ij} \in \mathbb{R}^{15}$. For each bounding box b_i, the normalized center coordinates, width, height, and rotation angle are computed as

$$g_i^{\text{bbox}} = \left[\frac{x_i}{w}, \frac{y_i}{h}, \frac{w_i}{w}, \frac{h_i}{h}, a_i, \frac{w_i h_i}{wh}\right] \tag{12}$$

where (x_i, y_i) are the center coordinates; (w_i, h_i) are the width and height, respectively; a_i is the rotation angle; and w and h are the image width and height, respectively. The pairwise geometric relations between b_i and b_j are calculated as

$$g_{ij}^{\text{pair}} = \left[\sqrt{\left(x_i - x_j\right)^2 + \left(y_i - y_j\right)^2}, \text{IoU}\big(b_i, b_j\big), \frac{w_i h_i}{w_j h_j}\right]. \tag{13}$$

The geometric feature vector is formed by concatenating the individual bounding box features and the pairwise relations. The final object-pair feature vector f_{ij}^{GCAM} is obtained by concatenating the visual feature vector f_{ij}^{visual} with the feature mapped from g_{ij} through an MLP

$$g_{ij} = \left[g_i^{\text{bbox}}, g_j^{\text{bbox}}, g_{ij}^{\text{pair}}\right] \tag{14}$$

$$f_{ij}^{\text{GCAM}} = \left[f_{ij}^{\text{visual}}, \text{MLP}\big(g_{ij}\big)\right]. \tag{15}$$

2) Object-Level Context-Aware Module: When there are N_{box} objects in the image, they do not exist in isolation but are inherently structured within a hierarchical object-level context. Semantic and spatial structural relations exist between objects, requiring higher order contextual reasoning beyond low-level visual cues. Relying solely on the pixel-level context of each object is insufficient. It is also essential to consider the global interactions among objects. Therefore, we designed the OCAM to capture object-level contextual information and enhance pairwise feature representations.

For each object b_i, OCAM constructs a comprehensive input feature vector $\boldsymbol{x} \in \mathbb{R}^{N_{\text{box}} \times d_h}$ by integrating visual, semantic, and spatial information, where d_h denotes the dimension of the embedded feature space. The semantic feature $\boldsymbol{e}$ is derived from class logits $\boldsymbol{l}$. Spatial information is encoded using the bounding box coordinates and dimensions normalized by the image size, transformed into an embedding $\boldsymbol{p}$ via a two-layer MLP. Then, the features are concatenated and projected to a low-dimensional space through a linear transformation

$$\boldsymbol{e} = \text{Softmax}(\boldsymbol{l}) \cdot \boldsymbol{W_e} \tag{16}$$

$$\boldsymbol{p} = \text{MLP}\big(\boldsymbol{g}^{\text{bbox}}\big) \tag{17}$$

$$\boldsymbol{x} = \text{Linear}\big(\big[\boldsymbol{f}, \boldsymbol{e}, \boldsymbol{p}\big]\big) \tag{18}$$

where $\boldsymbol{W_e}$ is a learned embedding matrix and $\boldsymbol{g} = \{g_i\}_{i=1}^{N_{\text{box}}}$ [g_i is defined in (12)]

To encode object-level context, the input features $\boldsymbol{x}$ are processed by a transformer encoder with sparse attention. Sparse attention focuses on the most relevant interactions, significantly reducing computational complexity compared to

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dense attention. The sparse scaled dot-product attention is computed as

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^\top}{\sqrt{d_k}} \cdot M_{\text{sparse}}\right)V \quad (19)$$

where $Q = XW_Q$, $K = XW_K$, and $V = XW_V$ are the query, key, and value matrices, respectively, with projection weights $W_Q, W_K, W_V \in \mathbb{R}^{d_h \times d_k}$, in which d_k is the dimension of the query and key embeddings. $M_{\text{sparse}} \in \{0,1\}^{N_{\text{box}} \times N_{\text{box}}}$ is the sparsity mask that determines the subset of key–value pairs attended to by each query. We use this transformer encoder based on a sparse attention mechanism (TESA) to embed object-level contextual information into x_i. After obtaining object representations $\boldsymbol{o} \in \mathbb{R}^{N_{\text{box}} \times d_h}$ that include object-level context, pairwise features f_{ij}^{OCAM} can be generated for each object pair

$$\boldsymbol{o} = \text{TESA}(\boldsymbol{x}) \quad (20)$$

$$f_{ij}^{\text{OCAM}} = [o_i, o_j]. \quad (21)$$

D. Relation Triplet Prediction

After obtaining pairwise features from the OCAM and GCAM, we can perform final feature fusion and relation prediction for each object pair. Specifically, the pairwise features f_{ij}^{OCAM} and f_{ij}^{GCAM} are concatenated and processed through an MLP for relation prediction and get the final relation triplet prediction. The predicted relations are then supervised using a cross-entropy loss function

$$\tilde{r_{ij}} = \sigma\left(\text{MLP}\left(\left[f_{ij}^{\text{GCAM}}, f_{ij}^{\text{OCAM}}\right]\right)\right) \quad (22)$$

$$\mathcal{L}_{\text{RTPM}} = -\frac{1}{|\tilde{\mathcal{P}}|}\sum_{(i,j)\in\tilde{\mathcal{P}}}\left[\tilde{y_{ij}}\log\left(\tilde{r_{ij}}\right) + \left(1-\tilde{y_{ij}}\right)\log\left(1-\tilde{r_{ij}}\right)\right] \quad (23)$$

where $\tilde{\mathcal{P}}$ is the set of the object pairs filtered by RopPN, $\tilde{y_{ij}}$ is the GT relation label for the pair between (b_i, b_j), and $|\tilde{\mathcal{P}}|$ is the total number of filtered pairs.

The overall loss of EGCAN is formulated as the sum of the losses from RopPN and RTPM

$$\mathcal{L}_{\text{EGCAN}} = \mathcal{L}_{\text{RopPN}} + \mathcal{L}_{\text{RTPM}}. \quad (24)$$

V. Experiments

A. Object Detection

1) Evaluation Tasks and Metrics: Current methods in SGG are mainly divided into one- and two-stage approaches, with the latter being more established and numerous. Object detection is a crucial component of the two-stage approaches in SGG. The ReCon1M dataset, with its extensive annotations of directed bounding boxes, allows for object detection tasks to be conducted. Models trained on this dataset can serve both as detector components for SGG tasks and can be independently evaluated for their object detection capabilities on the ReCon1M dataset. Object detection tasks primarily include two types: HBB detection and OBB detection. HBB detection, which uses axis-aligned rectangular boxes, is suitable for objects with regular shapes or where orientation is not crucial, offering simplicity and computational efficiency. In contrast, OBB detection utilizes rotated rectangular boxes to locate objects, which is more suitable for irregularly shaped or direction-specific objects, such as ships or airplanes, providing more precise position and shape information, but with a higher computational complexity. Although our dataset only has annotations for OBB, since OBB annotations can be directly converted into HBB annotations, we conduct both types of object detection tasks for benchmark testing on this dataset.

We use mean average precision (mAP) at an IoU threshold of 0.5 (mAP@50), introduced in the PASCAL VOC 2007 benchmark, as the evaluation metric. For HBB detection tasks, IoU is computed based on horizon bounding boxes, while for OBB detection tasks, IoU is calculated using OBBs to better capture the spatial alignment of objects with arbitrary orientations.

2) Baseline Models: For the two detection tasks, we selected four representative models for evaluation. In the HBB detection task, we chose Faster R-CNN, Cascade R-CNN, FCOS, and VFNet as baselines. For the OBB detection task, we selected Rotated Faster R-CNN, Oriented R-CNN, RoI Transformer, and ReDet as baselines.

3) Training Details: Our dataset contains very-high-resolution images (e.g., 6000 × 6000), which most baseline models cannot process directly as input. Since a considerable number of images in the dataset have at least one dimension (height or width) $\leq$ 1000, we retain these images at their original dimensions to avoid redundant cropping. For larger images, we divide them into 800 × 800 blocks with a stride of 400. For both HBB detection and OBB detection tasks, we use standard hyperparameters from models in mmdetection and mmrotate. Each model is trained on two RTX 3090 GPUs, with Adam as the optimizer and an initial learning rate of 0.005. We set the batch size to 4 and the initial learning rate to 1×10^{-4}. A stepwise learning rate schedule is employed, and the training is conducted for a total of 24 epochs.

4) Results and Analysis: In our study, we conduct eight distinct experiments using varied models, the results of which are shown in Table IV. This exploration reveals that the choice of model architecture plays a pivotal role in influencing the mAP metric. Specifically, in the HBB detection task, the FCOS model tops the chart with an mAP of 0.339, while the Cascade R-CNN achieved only 0.268, showing a difference of 0.071. In the OBB detection task, ReDet achieves an mAP score of 0.356, while Rotated Faster R-CNN achieves the lowest mAP score of 0.327, showing a relatively small difference in accuracy. In the OBB detection task, the mAP of ROI Transformer and ReDet reached 0.351 and 0.356, respectively, both surpassing the best-performing FCOS in the HBB detection task. Additionally, the mAP of Rotated Faster R-CNN and Faster R-CNN, which share the same model architecture, is 0.327 and 0.305, respectively, indicating that the same architecture learns more discriminative object features in the OBB detection task. Furthermore, as the OBB detection task uses rotated IoU for evaluation, which is stricter than horizontal IoU, the OBB detection model is considered more effective when the mAP values of HBB and OBB are the same.

Moreover, the substantial variance in AP scores across different classes, exacerbated by class imbalance, is characteristic of remote sensing imagery. Notably, in both the HBB

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TABLE IV
BASELINE RESULTS OF OBJECT DETECTION ON RECON1M

Category	HBB Results				OBB Results			
	Faster R-CNN [66]	Cascade R-CNN [67]	FCOS [68]	VFNet [69]	Rotated Faster R-CNN [66]	Oriented R-CNN [70]	ROI Transformer [71]	ReDet [72]
van	0.230	0.187	0.323	0.446	0.311	0.330	0.360	0.403
small car	0.283	0.247	0.328	0.451	0.336	0.354	0.414	0.426
building	0.325	0.304	0.333	0.362	0.315	0.309	0.316	0.317
road	0.373	0.297	0.284	0.411	0.186	0.242	0.230	0.258
airplane	0.907	0.906	0.908	0.907	0.906	0.908	0.908	0.907
block	0.541	0.454	0.474	0.520	0.441	0.444	0.444	0.452
parking lot	0.384	0.299	0.311	0.356	0.329	0.327	0.338	0.333
motorboat	0.500	0.424	0.544	0.546	0.550	0.619	0.574	0.621
dump truck	0.240	0.110	0.144	0.226	0.268	0.353	0.362	0.376
cargo truck	0.341	0.221	0.318	0.334	0.410	0.426	0.434	0.412
dry cargo ship	0.582	0.493	0.595	0.578	0.539	0.628	0.599	0.648
runway	0.517	0.446	0.448	0.510	0.256	0.301	0.334	0.352
container	0.302	0.240	0.260	0.303	0.150	0.224	0.241	0.208
water	0.645	0.607	0.600	0.666	0.616	0.620	0.633	0.640
intersection	0.618	0.492	0.628	0.631	0.621	0.619	0.625	0.626
fishing boat	0.293	0.193	0.345	0.289	0.262	0.312	0.286	0.283
storage tank	0.382	0.427	0.356	0.473	0.459	0.490	0.489	0.491
airport	0.759	0.726	0.572	0.727	0.714	0.703	0.712	0.738
...	...	...	...	...	...	...	...	...
stadium	0.189	0.139	0.263	0.110	0.219	0.204	0.226	0.286
locomotive	0.025	0.000	0.000	0.000	0.023	0.000	0.000	0.000
helicopter-apron	0.085	0.000	0.151	0.078	0.097	0.062	0.164	0.091
mAP	0.305	0.268	**0.339**	0.326	0.327	0.338	0.351	**0.356**

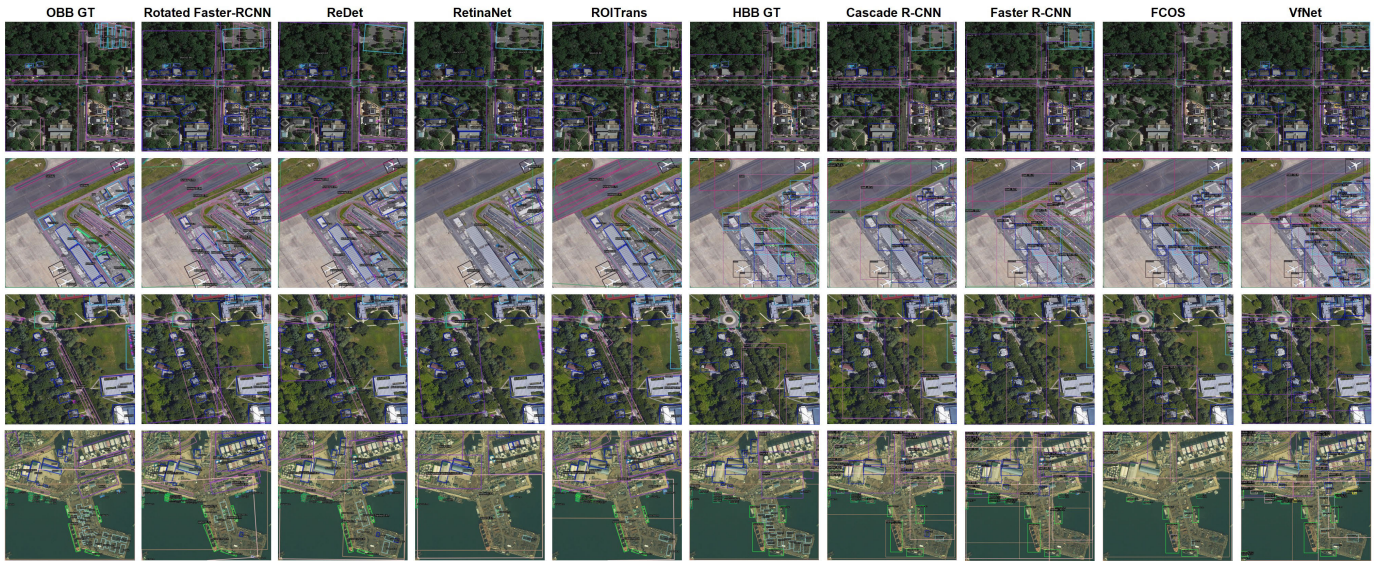


Fig. 10. Visualization of detection results.

and OBB detection tasks, aircraft objects demonstrate the highest AP scores, significantly surpassing other categories. This phenomenon likely results from their ample annotations and distinct textural and geometric features, which facilitate the detector's training on more discriminative characteristics, thereby boosting accuracy. Conversely, the least annotated categories in the dataset (dams, locomotives, and gas stations), which have fewer than 100 annotations each, suffer from insufficient training data. This lack of data prevents effective learning of discriminative features, resulting in zero AP scores across all experiments. The visualization of representative detection results is shown in Fig. 10, which further illustrates the performance differences among detection models and object categories.

B. Scene Graph Generation

1) Evaluation Tasks and Metrics: SGG focuses on identifying objects in images and determining the relations between

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TABLE V
BASELINE RESULTS OF SGG ON RECON1M. FPS IS MEASURED IN THE PREDCLS TASK EVALUATION

model	PredCls		SGCls		SGDet		FPS
	R@20/50/100/500	mR@20/50/100/500	R@20/50/100/500	mR@20/50/100/500	R@20/50/100/500	mR@20/50/100/500	
IMP [30]	57.8/72.0/80.2/90.4	30.3/41.7/49.6/61.4	38.9/46.4/50.1/53.7	15.9/21.6/25.3/29.3	7.0/10.6/13.6/20.0	1.2/1.7/2.2/3.7	10.8
MOTIFS [57]	61.3/77.1/86.2/95.1	41.2/55.0/64.7/78.5	39.4/47.2/51.0/54.3	17.7/23.9/27.6/31.2	15.0/19.5/22.7/28.6	4.3/5.9/7.4/10.7	11.5
VtransE [45]	60.7/76.4/85.5/95.0	34.7/47.9/57.2/71.6	38.5/46.2/50.0/53.4	16.5/22.1/25.4/29.4	14.0/18.7/21.9/27.7	4.2/5.7/6.9/9.9	12.1
RTN [64]	61.2/76.9/85.9/95.4	42.0/55.6/64.4/77.9	39.2/47.1/51.0/54.5	19.1/24.5/27.9/32.0	15.0/19.7/23.0/28.7	3.5/5.2/6.6/10.0	8.9
GPSNet [73]	61.0/76.7/85.9/95.5	26.7/51.2/61.2/77.4	38.9/46.6/50.5/54.0	16.9/23.4/26.8/30.6	10.3/13.4/15.8/22.4	2.0/2.7/3.4/6.2	11.2
GRCNN [74]	61.3/77.0/86.1/95.4	36.1/50.4/60.5/76.6	39.3/47.1/51.0/54.3	17.6/23.9/27.4/31.2	8.6/11.6/14.0/20.4	1.3/1.8/2.5/4.5	0.3
SQUAT [75]	59.3/74.2/82.8/93.6	38.6/51.7/60.5/75.7	37.2/44.6/48.5/52.8	15.0/20.0/23.7/28.0	13.5/17.3/19.8/24.7	3.3/4.3/5.2/7.3	8.6
BGNN [76]	61.0/76.3/85.3/95.2	37.8/51.0/60.6/76.4	35.8/42.9/47.0/51.5	13.6/18.8/22.0/25.8	15.1/16.9/17.7/18.3	**8.5/10.5/11.4/12.8**	4.3
PENet [77]	60.8/76.2/85.2/94.9	**45.0/57.1/66.2**/78.6	38.2/45.7/49.6/53.2	15.4/20.8/24.5/28.2	12.3/13.8/14.5/15.0	5.7/7.3/8.1/8.9	9.5
Motifs+DPL [78]	55.2/69.0/77.3/88.1	15.5/21.4/25.4/31.9	36.7/43.8/47.6/52.1	8.4/11.6/13.7/16.8	10.3/11.6/12.3/12.9	3.0/3.7/4.1/4.3	11.2
EGCAN	**62.2/78.0/87.0/95.6**	42.5/56.4/65.8/**78.8**	**40.2/48.2/52.0/55.3**	**19.6/26.4/30.2/34.0**	**16.5/21.5/24.7/30.0**	4.6/6.4/7.8/10.7	**13.0**

them. To comprehensively assess the capabilities of our dataset and the effectiveness of various algorithms, we have segmented the SGG task into three distinct subtasks: predicate classification (PreCls), scene graph classification (SGCls), and scene graph detection (SGDet). PreCls involves predicting the relations between pairs of objects given their labels and bounding boxes. The primary challenge here is to accurately classify the type of relation without misinterpreting the visual and spatial cues. In SGCls, both the object labels and their relations need to be classified, assuming that the bounding boxes of objects are known. This tests the algorithm's ability to integrate object recognition with relation prediction. SGDet is the most comprehensive subtask, where the algorithm must detect objects, classify them, and predict their interrelations, starting from raw images. It simulates a real-world scenario where both objects and their relations need to be inferred without prior knowledge. The current methods in SGG can also be divided into one- and two-stage methods. Compared with the vigorous development of two-stage methods, one-stage methods are still in the initial stage. Two-stage methods typically achieve higher accuracy in both object detection and relation prediction compared to one-stage methods. Additionally, many existing benchmark datasets and evaluation metrics in the field of SGG are designed with two-stage methods in mind. Primarily focusing on two-stage methods can facilitate a more meaningful evaluation and comparison of results.

In the relation detection task, we have opted to use mean recall at K (mR@K) as the evaluation metric for model performance. Recall is defined as the proportion of actual relation triplets successfully detected by the model out of the total real relation triplets. It is calculated as the number of true positives (TPs) divided by the sum of TPs and false negatives (FNs)

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \tag{25}$$

Recall@K is an extended metric of recall, where K represents the number of top highest scoring predictions retained for calculating recall, focusing specifically on the successful detection of real relation triplets within these predictions. This approach allows researchers to select different K values based on varying performance assessment needs. In experiments with natural scene datasets, the K value is typically set to 20, 50, or 100. However, due to the significantly higher annotation density of ReCon1M compared to natural scene datasets, we additionally set the K value to 500. mR@K is the mean of the Recall@K values across various relation categories. In the SGDet task, the maximum number of object bounding boxes generated by the detector is fixed, whereas the PredCls task utilizes GT boxes for relation prediction. In ReCon1M, the number of object bounding boxes per cropped image can exceed 1000, resulting in over one million potential object pairs. Hence, we use frames per second (FPS) as a key metric to evaluate the efficiency of each model in performing relation prediction under scenarios with dense objects, as tested in the PredCls task.

2) Baseline Models: We have selected ten two-stage models in SGG, along with the proposed EGCAN, for evaluation. These models include IMP, Motifs, RTN, VTransE, BGNN, GPS-Net, PE-Net, Motifs + DPL, Graph R-CNN, and SQUAT. These models include both classic and the latest SOTA models, which can effectively serve as benchmarks for the SGG task on our dataset.

3) Training Details: Due to the two-stage approach of SGG methods, which separates object detection and relation prediction into two parts, we need to use a separately trained object detection model as a detector. For the sake of convenience in comparison with previous research, we have chosen to use the widely adopted Rotated Faster R-CNN as the detector. The image input settings for the detector are the same as those in the object detection experiments. When training the SGG model, we need to first load the weights of the previously trained detector and freeze the detector's parameters, training and updating only the parameters of the relation detector. For each task of each model, we train on a single RTX3090, using

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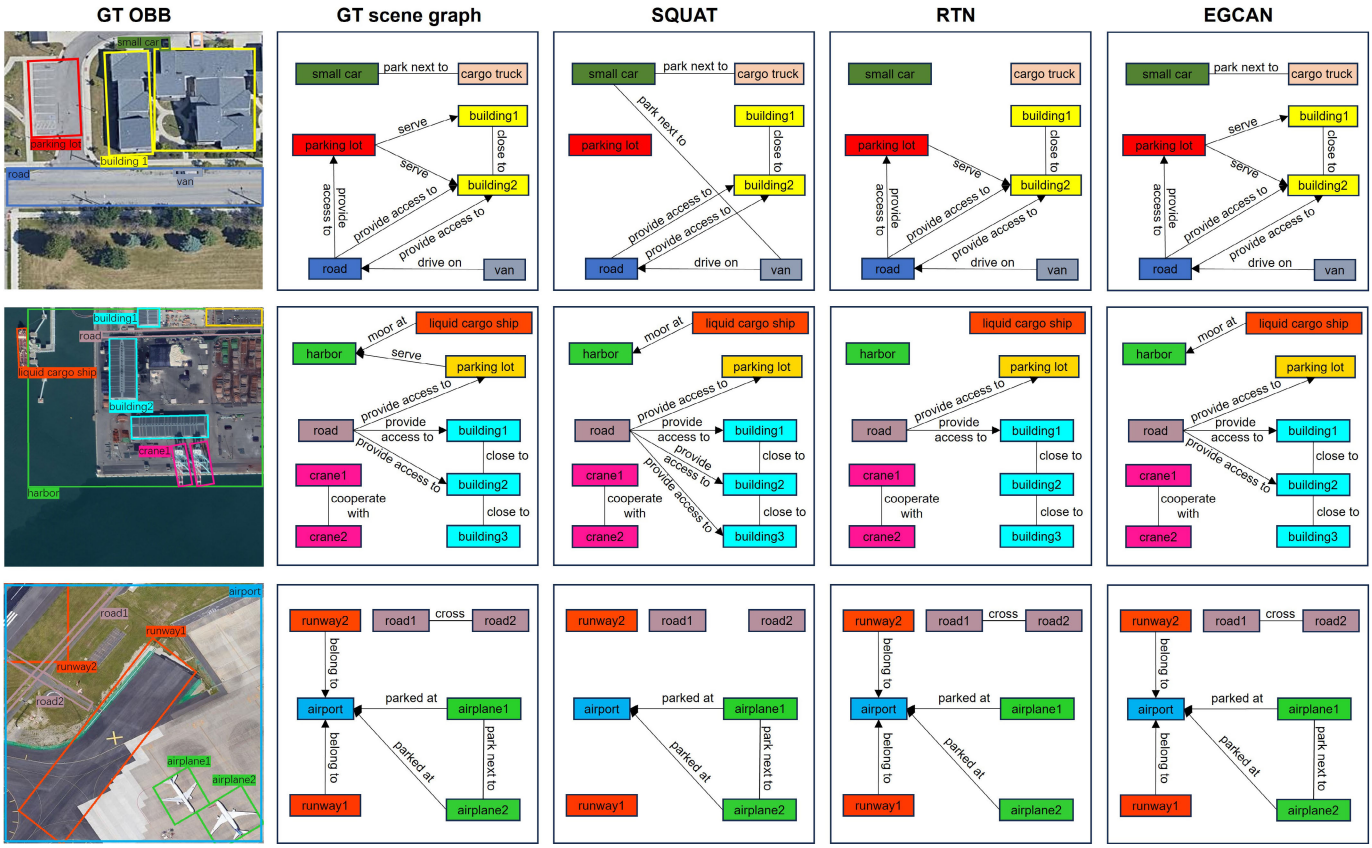


Fig. 11. Visualization results of three models (SQUAT, RTN, and EGCAN). The three sets of visualizations from left to right are the image annotated with OBB, the GT scene graph, the scene graph predicted by SQUAT, the scene graph predicted by RTN, and the scene graph predicted by EGCAN.

Adam as the optimizer with an initial learning rate of 1×10^{-4} and employing cosine annealing as the learning rate scheduling strategy, for a total of 24 epochs.

4) Results and Analysis: We conduct extensive experiments on the meaningful relation prediction task, and the results are reported in Table V. The widely used and classic models, such as IMP, Motifs, and PENet, can produce effective values of recall and mean recall. This demonstrates the effectiveness of our dataset while also showing the significant challenges that exist in the proposed dataset. PENet effectively addresses the relation prediction problem through a prototype-representation matching approach. This method significantly improves the accuracy of relation prediction, as evidenced by its best performance on the PredCls task, achieving mR@20/mR@50/mR@100 of 45.0%/57.1%/66.2%. However, its performance on the R@K metrics is generally average. BGNN achieves the best mR@K results on the SGDet task, with mR@20/50/100/500 reaching 8.5%/10.5%/11.4%/12.8%. However, its performance on other metrics is less than satisfactory. These improvements can be attributed to BGNN's bipartite graph structure and heterogeneous information propagation mechanism, which allow the model to effectively capture the complex interactions between object nodes and relation nodes. Nevertheless, its performance on other metrics remains suboptimal.

In addition, there may be a lot of room for improvement because existing methods are not well adapted to model remote sensing scenarios. To tackle the issue of dense objects and relations in remote sensing images, the proposed EGCAN introduces RopPN, an object-pair selection module based on feature fusion and self-attention enhancement mechanisms. During inference, RopPN reduces computational overhead by pre-screening and filtering out background object pairs, thus significantly improving inference efficiency when handling dense object pairs. Table V demonstrates that EGCAN achieves an FPS of 13.0, notably outperforming other models, thereby highlighting its efficient inference capability.

Remote sensing images are characterized by complex scenes, large image sizes, and objects with significant scale variations, and relation prediction is highly dependent on the features of the objects and the interactions between them. EGCAN introduces OCAM and GCAM, which fuse visual, geometric, and semantic features between objects to generate fine-grained pairwise features embedded with object-level contextual information for relation prediction. In the 24 metrics across the three SGG subtasks listed in Table V, EGCAN has achieved the best performance in 17 metrics. Specifically, in the PredCls task, R@20/50/100/500 has reached 62.2%/78.0%/87.0%/95.6%; in the SGCls task, R@20/50/100/500 has reached 40.2%/48.2%/52.0%/55.3%; and in the SGDet task, R@20/50/100/500 has reached 16.5%/21.5%/24.7%/30.0%. EGCAN demonstrates the optimal performance in Recall@K across all three subtasks, indicating its strong ability to capture object relations,

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TABLE VI
COMPARISON BETWEEN HBB DETECTOR AND OBB DETECTOR ON EGCAN. THE DETECTORS USED IN BOTH CASES ARE FASTER R-CNN

Task	HBB		OBB	
	R@20/50/100/500	mR@20/50/100/500	R@20/50/100/500	mR@20/50/100/500
PredCls	61.3/76.7/85.6/95.0	41.0/53.2/62.8/76.2	**62.2/78.0/87.0/95.6**	**42.5/56.4/65.8/78.8**
SGCls	38.9/45.6/50.3/53.9	15.2/24.0/26.9/32.5	**40.2/48.2/52.0/55.3**	**19.6/26.4/30.2/34.0**
SGDet	**17.0**/20.8/23.5/28.5	**5.1**/6.2/7.2/8.7	16.5/**21.5/24.7/30.0**	4.6/**6.4/7.8/10.7**

TABLE VII
COMPARISON WITH DIFFERENT VALUES OF N FOR OBJECT PAIR SELECTION AND ABLATION STUDIES IN SGDET TASK. "–" INDICATES THE NONAPPLICATION OF ROPPN

N	R@20/50/100/500	mR@20/50/100/500	FPS
500	**16.8/21.7/24.7**/27.6	**4.7/6.4/7.9**/9.5	**10.5**
1000	16.5/21.5/**24.7/30.0**	4.6/**6.4**/7.8/**10.7**	9.3
2000	15.7/20.7/24.0/29.8	4.3/6.1/7.6/10.8	8.0
5000	14.7/19.6/22.9/28.8	4.0/5.9/7.4/10.5	4.5
-	15.0/19.8/23.4/29.2	4.0/6.0/7.5/10.1	2.7

TABLE VIII
ABLATION STUDY ON DIFFERENT PAIRWISE FEATURE GENERATION STRATEGIES. "CONCAT" INDICATES THE CONCATENATION OF ONLY THE VISUAL FEATURES OF THE SUBJECT BOUNDING BOX AND OBJECT BOUNDING BOX

Strategy	R@20/50/100/500	mR@20/50/100/500
OCAM+GCAM	**16.5/21.5/24.7/30.0**	**4.6/6.4/7.8/10.7**
OCAM	14.1/19.5/21.2/26.3	3.0/4.7/5.4/7.1
GCAM	5.0/7.2/7.9/8.1	0.5/0.7/1.0/1.3
concat	1.7/2.3/2.8/4.9	0.1/0.1/0.2/0.3

especially excelling at predicting high-frequency relations. This ensures that the most relevant relations are correctly identified, leading to the generation of comprehensive scene graphs with strong generalization capability.

EGCAN has achieved 95.6% in mR@500 for PredCls and 19.6%/26.4%/30.2%/34.0% in mR@20/50/100/500 for SGCls, reaching the highest performance in these five mR@K metrics. However, in the PredCls task, EGCAN's performance in mR@K is outperformed by PENet, and in the more challenging SGDet task, its performance is also inferior to BGNN's. The ReCon1M dataset exhibits a pronounced long-tail distribution in the instance counts of relation categories. EGCAN's suboptimal performance on mR@K metrics indicates that while the model effectively identifies high-frequency relations, its ability to handle more complex or low-frequency relations remains limited. This highlights an important area for improvement. Future research could focus on enhancing the model's capacity to process low-frequency relations, thereby addressing this limitation.

We select three models (SQUAT, RTN, and EGCAN) for SGG visualization. As illustrated in Fig. 11, SQUAT exhibits noticeable mispredictions and omissions. For instance, in the first set of visualized results, SQUAT erroneously predicts a "park next to" relation between the small car and the van that are spatially distant, and it fails to detect the "provide access to" relation between road and parking lot. RTN generates more accurate scene graphs than SQUAT in the visualizations. Although no incorrect relations are observed in the given examples, RTN still exhibits omissions. Specifically, in the first set of visualized results, RTN fails to infer the "park next to" relation between the small car and the cargo truck. Similarly, in the third set of visualized results, RTN fails to infer the "park next to" relation between airplane1 and airplane2. In contrast, our EGCAN generates markedly more accurate scene graph predictions across all three sets of examples. It only misses a single relation in the second set of visualized results, while all other relations are inferred completely and correctly.

5) Ablation Study on the Proposed Method: We perform ablation studies and comparative experiments on the key modules of EGCAN to analyze their contributions to the model's performance.

a) Effect of detector type (HBB and OBB): From Table IV, it can be observed that when experiments are conducted on OBB and HBB detection tasks using the same model architecture, the mAP of Rotated Faster R-CNN is 0.327, which is higher than the 0.305 achieved by Faster R-CNN. Under the same model architecture, a higher mAP indicates that the model learns more discriminative object features. We conduct comparative experiments on SGG tasks using HBB-based EGCAN (HBB-EGCAN) and OBB-based EGCAN (OBB-EGCAN) on the ReCon1M dataset, with the results presented in Table VI. The experimental results show that in the PredCls and SGCls tasks, HBB-EGCAN demonstrates significant accuracy advantages, outperforming OBB-EGCAN in all metrics. In the SGDet task, HBB-EGCAN achieves higher R@20 and mR@20 scores than OBB-EGCAN but performs worse on other metrics.

These results can be attributed to two main factors: the inherent advantages of OBB detectors and the reliance of PredCls and SGCls tasks on GT boxes. In certain test images, the large number of densely distributed GT boxes leads to significant overlap and redundant noise when OBB annotations are converted to HBB, reducing the model's relation prediction effectiveness. In contrast, for the SGDet task, where

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the number of predicted boxes is capped, the performance gap between HBB-EGCAN and OBB-EGCAN narrows. This suggests that under sparsely distributed object conditions, HBB-based SGG models can achieve comparable or even slightly better performance than OBB-based models. Nevertheless, across 24 evaluated metrics, OBB-EGCAN outperforms HBB-EGCAN in 22 metrics, underscoring the superiority of SGG tasks based on OBB detectors over those based on HBB detectors.

b) Effect of RopPN: RopPN is designed to pre-screen dense object pairs by generating relevance scores for each pair and retaining the top N pairs for subsequent relation prediction. Our primary objective is to investigate how variations in the parameter N affect the overall performance of the model. To this end, we conduct ablation studies on RopPN under the most challenging SGDet task. We experiment with N values of 500, 1000, 2000, and 5000 to assess the impact of different N values on the model performance. As shown in Table VII, FPS decreases significantly as N increases, indicating that smaller N values lead to higher inference efficiency. We observe that when $N = 500$, the model achieved the best performance in R@20/50/100 and mR@20/50/100, slightly outperforming $N = 1000$ and significantly surpassing other N values. However, the model's performance at R@500 and mR@500 is noticeably worse for all N values, representing a notable limitation. When $N = 1000$, the model's accuracy and speed metrics are superior to those at $N = 2000$ and $N = 5000$, making $N = 1000$ the optimal choice for the best overall performance. This suggests that, as N increases, the model's performance improves initially but eventually declines, and the choice of N has a significant impact on the overall performance. Furthermore, in the ablation study of RopPN, the results in Table VII show that removing RopPN yields similar R@K and mR@K accuracies to those at $N = 5000$, but with a significantly lower FPS of 2.7 compared to 4.7 at $N = 5000$, and much lower than FPS at other smaller N values. This highlights the importance of RopPN, as it enhances both model accuracy and inference efficiency.

c) Effect of OCAM and GCAM: OCAM and GCAM are fundamentally designed to extract more discriminative pairwise features for relation prediction. To evaluate their effectiveness, we conduct three sets of ablation experiments, comparing their individual and combined impacts. When both OCAM and GCAM are removed, a baseline strategy is adopted by simply concatenating the visual features of the subject and object bounding boxes. As shown in Table VIII, the combined use of OCAM and GCAM for pairwise feature generation significantly outperforms the other three strategies. Using OCAM alone results in a performance drop, though the decrease is smaller than when only GCAM is used, suggesting that object-level contextual information and interaction are essential. In contrast, relying solely on the concat strategy for pairwise feature generation leads to a substantial performance decline, underscoring the importance of extracting discriminative pairwise features as a foundation for effective relation prediction.

VI. Conclusion

We present ReCon1M, the first large-scale, million-level relation benchmark dataset specifically designed for relation comprehension in remote sensing imagery. This dataset not only addresses the scarcity of specialized data for this domain but also enhances our understanding of complex visual scenes through detailed annotations of objects and their interrelationships. With its comprehensive coverage of diverse object and relation categories, ReCon1M is poised to facilitate a deeper understanding of remote sensing imagery and promote further innovation in SGG technology. Furthermore, the proposed EGCAN demonstrates superior performance across multiple accuracy metrics, showcasing its potential for practical applications in relation prediction tasks. ReCon1M not only enhances the generalization capabilities of current algorithms but also encourages the development of novel methods tailored to the unique challenges of remote sensing imagery. As demonstrated through extensive experiments, this dataset promotes more nuanced and efficient object relation comprehension, laying the groundwork for future research in remote sensing, including urban planning, disaster management, environmental monitoring, and other applications. We hope that this contribution inspires further innovations and fosters deeper integration of artificial intelligence in remote sensing applications.

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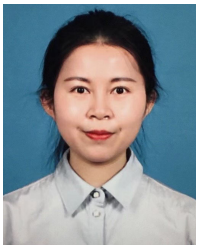
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